

Session 5 · Deck 5.A3

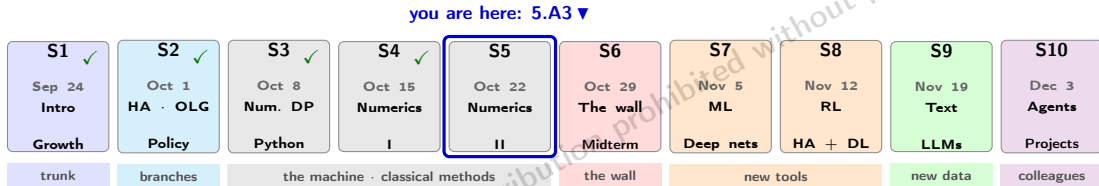
First Order: Four Equations, Four Unknowns

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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 5.A1 what perturbation computes · 5.A2 the model and the parameter · **5.A3** first order · 5.A4 linear RE systems · 5.A5 higher order, risk, pruning · 5.A6 the worked example.

Packaging Everything Into One Object

$$F(k_t, z_t; \chi) = \begin{pmatrix} \frac{1}{c(k_t, z_t; \chi)} - \beta \frac{\alpha e^{\rho z_t + \chi \sigma \varepsilon_{t+1}} (k(k_t, z_t; \chi))^{\alpha-1}}{c(k(k_t, z_t; \chi), \rho z_t + \chi \sigma \varepsilon_{t+1}; \chi)} \\ c(k_t, z_t; \chi) + k(k_t, z_t; \chi) - e^{z_t} k_t^\alpha \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Why bother

One object, two components — Euler equation and resource constraint — so that “differentiate and match” becomes mechanical rather than a case-by-case argument.

Write H_i for the partial derivative of the Euler component with respect to its i -th argument. The evaluation point $(k^*, 0; 0)$ is dropped when it is clear.

Quant _ Macro Lec. 9, “Defining $F(k_t, z_t; \lambda) = 0$ (1/2)”.

An Identity, Not an Equation

Once the *true* policies are substituted in, $F(k_t, z_t; \chi) = 0$ holds for **every** (k_t, z_t, χ) in a neighbourhood. It is an identity.

And a function that is identically zero has identically zero derivatives:

$$F \equiv 0 \implies F_k = F_z = F_\chi = F_{kk} = \dots = 0.$$

The unknown Taylor coefficients $\{c_k, c_z, k_k, k_z, \dots\}$ appear *inside* those derivatives, through the chain rule. Setting each derivative to zero therefore produces **equations for them**.

First order needs (F_k, F_z, F_χ) . Second order also needs $(F_{kk}, F_{kz}, F_{zz}, \dots)$, and so on.

Quant_Macro Lec. 9, "Why Derivatives Pin Down Unknowns" and "First-Order Derivatives of F ".

This single observation — *identity, therefore derivatives vanish* — is where every perturbation equation in this course comes from.

It holds only where the true policy is differentiable. A binding constraint breaks the identity's smoothness, not the logic.

Why Doesn't $F = 0$ Itself Give Us the Equations?

A fair question: $F = 0$ has two components. Why are those not two equations for the first-order coefficients?

Because $F = 0$ is the zeroth-order condition. At the expansion point $\hat{k}_t = 0, z_t = 0, \chi = 0$, so every linear term in the policy expansion — $c_k \hat{k}_t, c_z z_t, c_\chi \chi$ — multiplies zero and drops out. What survives pins down the **steady-state levels**, which we already knew.

The slopes come from matching the coefficients on $\hat{k}_t, z_t$ and χ , which is the same thing as imposing

$$F_k = 0, \quad F_z = 0, \quad F_\chi = 0.$$

Quant_Macro Lec. 9, "Why Don't We Count $F = 0$ Itself as Equations?".

Contrast with projection

Block B does the opposite: it imposes $F = 0$ at **many points** (k_i, z_i) , so the unknown coefficients appear directly in the equations and no derivative is ever taken. One method differentiates once and evaluates nowhere; the other evaluates everywhere and differentiates nothing.

The First-Order Conditions

Six unknowns enter at linear order: $\{c_k, c_z, c_\chi, k_k, k_z, k_\chi\}$. Differentiating $F = 0$ and applying the chain rule gives, schematically,

$$F_k(k^*, 0; 0) = H_1 c_k + H_2 c_k k_k + H_3 + H_4 k_k = 0$$

$$F_z(k^*, 0; 0) = H_1 c_z + H_2 (c_k k_z + c_z \rho) + H_4 k_z + H_5 = 0$$

$$F_\chi(k^*, 0; 0) = H_1 c_\chi + H_2 (c_k k_\chi + c_\chi) + H_4 k_\chi + H_6 = 0$$

Each line is a 2-vector

F has two components, so $F_k = 0$ is two scalar equations, and so is $F_z = 0$.

Note F_k is **quadratic** in the unknowns: the term $H_2 c_k k_k$ is a product of two of them. First order is a linear approximation, not a linear system.

Quant_Macro Lec. 9, "First-order approximation". The quadratic term is why 5.A4 needs a matrix quadratic and a root selection.

Six Unknowns, but Two Disappear

Count the restrictions. Differentiating with respect to the two *state* variables gives

$$F_k = 0 \Rightarrow 2 \text{ equations}, \quad F_z = 0 \Rightarrow 2 \text{ equations},$$

so four scalar equations for $\{c_k, c_z, k_k, k_z\}$ — and nothing yet for c_χ, k_χ .

Those two are handled separately, by

$$F_\chi = H_1 c_\chi + H_2 (c_k k_\chi + c_\chi) + H_4 k_\chi + H_6 = 0.$$

This is **linear** in (c_χ, k_χ) . And if it is also **homogeneous** — if $H_6 = 0$ — then the only solution is

$$c_\chi = k_\chi = 0.$$

Quant_Macro Lec. 9, "First-Order Approximation: Six Unknowns, but Two Disappear".

Certainty equivalence

$c_\chi = k_\chi = 0$ means the shock scale enters **no first-order coefficient**. To a linear approximation, an agent facing risk behaves exactly like one who does not.

So there is no precautionary saving at first order — not because the model lacks it, but because the approximation cannot see it.

Why $H_6 = 0$

H_6 is whatever sits in F_χ without multiplying c_χ or k_χ — the constant term. It is the direct effect of χ on the equilibrium conditions, holding the policies fixed.

At the expansion point $\chi = 0$, and the innovation enters as $\chi\sigma\varepsilon_{t+1}$ inside an expectation with

$$\mathbb{E}_t[\varepsilon_{t+1}] = 0.$$

A *linear* change in the shock scale therefore has *no* first-order effect: the term drops out.

With no constant, the equation in (c_χ, k_χ) is homogeneous, and the unique non-degenerate solution is zero.

Quant_Macro Lec. 9, "Why $H_6 = 0$ and the System is Homogeneous".

Where risk survives

Risk enters through the **variance**, which is quadratic in χ . So it first appears in $c_{\chi\chi}$ — a second-order term. 5.A5.

$\mathbb{E}[\varepsilon] = 0$ is doing the work. Change the shock to something with a non-zero mean effect at first order and $H_6 \neq 0$.

Exactly Four Equations for Exactly Four Unknowns

Collecting:

Source	Equations
$F_k = 0$, both components	2
$F_z = 0$, both components	2
$F_\chi = 0 \Rightarrow c_\chi = k_\chi = 0$	—
Total	4
Unknowns c_k, c_z, k_k, k_z	4

Square, and generically uniquely solvable. Higher-order coefficients need higher derivatives of F , and each order is linear given the ones below.

Quant_Macro Lec. 9, "Summary: Why Exactly Four Equations for Four Unknowns".

"Generically"

The counting says the system is square. It does not say a *stable* solution exists, or that only one does — the system is quadratic, so it can have two roots, and only one of them is economically admissible.

Choosing between them is what Blanchard–Kahn does, and it is the only genuinely new idea in Block A.

Measured: The Four Numbers, Computed Two Ways

On the quarterly calibration — $\alpha = 0.33$, $\beta = 0.99$, $\delta = 0.025$, $\rho = 0.95$ — with $k^* = 28.3484$, the four coefficients in log deviations are:

$\hat{k}'$ on $\hat{k}$	0.962061
$\hat{k}'$ on z	0.080097
$\hat{c}$ on $\hat{k}$	0.590408
$\hat{c}$ on z	0.322850

Read the first one: capital is **very** persistent, 0.962 per quarter, so a shock takes years to unwind.

Ledger pert_first_order: qz_vs_matching_max_abs_diff = 4.36×10^{-11} .

Two routes, one answer

Computed by generalized Schur on the linearized system (5.A4), and independently by root-finding on $F = 0$, $F_k = 0$, $F_z = 0$ with numerical derivatives. The two agree to 4.4×10^{-11} .

Two implementations of the same object, disagreeing nowhere. That is the check this course keeps asking for, and it is cheap here.

Measured: Certainty Equivalence, and What It Hides

c_χ and k_χ are the χ -derivatives of 5.A2's expansion — how much the policy shifts when the shock is switched on. Solved at $\sigma = 0$ they come out at

$$c_\chi = -2.7 \times 10^{-14}, \quad k_\chi = 3.4 \times 10^{-16},$$

i.e. zero. So far, so predicted.

The sharper test: add the *whole second-order block* and re-solve with risk on. If certainty equivalence is real, the four first-order coefficients should not move.

They move by 1.5×10^{-5} — against a coefficient of 0.59, that is one part in 40,000.

What does move is the **constant**: consumption shifts by -0.00041% of steady state. Small — but it is the entire economics of precaution, and first order reports it as exactly zero.

`Ledger pert_first_order.certainty_equivalence and higher_order.first_order_coeff_shift.`

In practice

A first-order solution describes *dynamics* well and *risk* not at all: welfare costs of uncertainty, risk premia and precautionary saving are identically zero in it.

Asset-pricing papers that report a zero equity premium from a linearized model are not finding an economic result.

Why We Still Need Blanchard–Kahn

The first-order solution is a pair of linear rules,

$$\hat{c}_t = \varphi_k \hat{k}_t + \varphi_z z_t, \quad \hat{k}_{t+1} = \psi_k \hat{k}_t + \psi_z z_t,$$

and substituting them into the equilibrium conditions leaves a **linear rational-expectations system**.

Such a system can have

- no stable solution,
- many stable solutions (indeterminacy),
- or exactly one.

Nothing in the counting argument distinguishes these. The quadratic system of the previous slides has more than one root, and only one of them gives bounded paths.

The coefficient count tells you the system is square. **Blanchard–Kahn tells you which root to keep** — and whether keeping one is possible at all.

1.B1 already used this logic informally: one stable and one unstable root, so from any k_0 there is a unique c_0 on the saddle path.

Quant _ Macro Lec. 9, “Why We Need Blanchard–Kahn After First-Order Perturbation”.

Takeaways

- Package the equilibrium conditions as $F(k, z; \chi) = 0$. Because the true policies make it an **identity**, every derivative of F is zero — and those are the equations.
- $F = 0$ itself is the *zeroth*-order condition: it pins the steady state, not the slopes. Projection (Block B) uses $F = 0$ instead, at many points.
- Six first-order unknowns, four equations from (F_k, F_z) , and $F_\chi = 0$ forcing $c_\chi = k_\chi = 0$ because $\mathbb{E}[\varepsilon] = 0$ kills the constant H_6 .
- That is **certainty equivalence**: risk enters no first-order coefficient, so precaution, risk premia and welfare costs of uncertainty are all exactly zero.
- Measured on the quarterly model: $\hat{k}'$ loads **0.962** on $\hat{k}$ and 0.080 on z ; two independent solution routes agree to 4.4×10^{-11} .
- Adding the whole second-order block moves those four coefficients by 1.5×10^{-5} and shifts the constant by -0.00041% — small, and the only place the economics of risk lives.

Bridge: Which Root, and Is There One?

We have a square system with a quadratic term, so it has more than one solution, and only one of them is economically meaningful. Picking it is not a numerical detail — it is the difference between a model with a unique equilibrium and one with a continuum.

The next deck does this properly: log-linearization and undetermined coefficients, the matrix quadratic, the generalized Schur decomposition that solves it, and the eigenvalue count that decides the case.

Next (5.A4)

Blanchard–Kahn counting; Klein's QZ solution; determinacy, indeterminacy and non-existence, measured on a New Keynesian model where the Taylor principle decides which one you get. And where Dynare fits.